Psi_Sat Stage 5 — Wiring Guide (battery deck)

Board: Stage 5 pack deck (EasyEDA Pro) · For Muhammad · Updated 2026-09-20
How to read this: Only what we walked through and confirmed is marked done. Open items are listed clearly. Stage 4 left/right header maps are the stack standard — Stage 5 matches them.

1. What Stage 5 is for

Stage 5 holds the **2S LiFePO■ pack** and a **2S pack protector**. It sends protected pack power up the stack on **VBAT_CHG** / **GND** so Stage 4's BQ25713 can charge and manage the system.

It is **not** the radio board anymore. The RFM98 was removed (battery noise). Radio moves to Stage 3 later.

2. Big picture (simple flow)

Cell B (+) → Cell mid → Cell A (–) → HY2122 + two AO3400A FETs → header GND (1–2). Cell B (+) also goes straight to header VBAT_CHG (9–10). Stack pins carry that into Stage 4.

3. What we removed (on purpose)

- RFM98-433 transceiver and all SPI / RADIO_* / ANT nets from this board
- Mid-board 1×2 male jumper (no mate on Stage 4)
- Old IRLML2502 + 1 kΩ on the pack (that path would dump Cell B; pin 38 was not a Stage 4 standard net)

4. Left 2×20 header (matches Stage 4)

Pin(s)	Net	On Stage 5
1–2	GND	Protected pack return (P–) after the two protect FETs.
3–4	+3V3_SYS	Pass-through name only (from Stage 4).
5–6	+5V0_SYS	Pass-through name only.
7–8	VSYS	Pass-through name only.
9–10	VBAT_CHG	Pack + from Cell B+ into Stage 4 charger path.
11–12	I2C SDA / SCL	Pass-through name only.
13	BURN_CTRL	Pass-through name only.
14+	NC	Unused.

5. Right 2×20 header (matches Stage 4)

Pin(s)	Net	On Stage 5
1–2	GND	Named to match Stage 4.
3–4	V_SOLAR_POS	Named to match Stage 4 (solar lives on Stage 4).
7	BURN_OUT	Named to match Stage 4.
Other	NC	Including old pin 38 — cleared.

6. Battery pack (2S series)

Item	Connection	Why

Cell B +	Left hdr 9–10 VBAT_CHG	Pack positive to Stage 4.
Cell B –	Cell A + (series mid)	2S string.
Cell A –	Protect FET side (VSS / M1 drain)	Raw pack – — not direct to hdr 1–2.
Holders	Keystone 1042 clips	On board now; retention still weak for flight (see §9).

7. Pack protection (HY2122) — done

IC: **HY2122 EB1A** (LCSC **C458515**), SOT-23-6, 2S LiFePO■ protector. Note: EB1A OV is ~3.95 V/cell; AB1A (~3.65 V) is a better LiFePO■ fit if you can source it later. BQ25713 on Stage 4 must still be set for 2S LiFePO■ charge limits.

Part	LCSC / MPN	Role
HY2122 EB1A	C458515	2S protect IC
M1, M2 AO3400A	C347475	Discharge + charge N-FETs (low side)
R1, R2 330 Ω	C23138 / 0603WAF3300T5E	VDD and VC series resistors
R3 2 kΩ	C22975 / 0603WAF2001T5E	CS sense resistor
C1, C2 0.1 µF	C14663 / CC0603KRX7R9BB104	Decouple VDD and VC to VSS

HY2122 pin	Name	Wired to
1	OD	M1 Gate (discharge FET)
2	OC	M2 Gate (charge FET)
3	CS	R3 → hdr 1–2 (GND / P–)
4	VC	R2 ← series mid (Cell B– / Cell A+)
5	VDD	R1 ← Cell B+ / VBAT_CHG
6	VSS	Cell A– (raw pack –)

FET	Role	Wiring (AO3400A: 1=G, 2=S, 3=D)
M1	Discharge	D→Cell A– / VSS; G→OD; S→M2 Source
M2	Charge	D→hdr 1–2 GND; G→OC; S→M1 Source

BQ25713 on Stage 4 is the charger. HY2122 is the pack safety cut. They do different jobs.

8. How voltage reaches Stage 4

No special mid jumper. Male stack pins on Stage 5 plug into Stage 4:

- Pack + on pins 9–10 (`VBAT_CHG`) → Stage 4 sense / charge path
- Pack – on pins 1–2 (`GND`) → Stage 4 ground (after protect FETs)

9. Still left (not done on Stage 5 wiring)

Item	Status	Simple next action
Cell retention	Open	Clips alone are weak for vibe/launch. Add strap, frame, or better holder + mounting holes.
Radio (RFM98)	Moved off Stage 5	Put transceiver on Stage 3 next to STM32 / SPI. Antenna stays Stage 6 later.
HY2122 AB1A swap	Optional	Better OV for LiFePO■ if you find stock; same footprint/wiring.
BQ25713 2S LiFePO■ setup	Stage 4 / firmware	Set charge voltage for 2S LFP so it matches the protector.
Flight battery process	Later	LSP-style testing, containment, inhibits — beyond this PCB wiring guide.

10. Quick checklist (as built)

- Radio deleted from Stage 5
- Left + right headers match Stage 4 names

- 2S series: B+ → mid → A–
- HY2122 + R1/R2/R3 + C1/C2 + M1/M2 wired
- Hdr 9–10 = Cell B+; hdr 1–2 = after FETs
- Cell A– not shorted straight to hdr 1–2

Pair with PsiSat_Stage4_Wiring_Guide.pdf. Refresh this file if retention or Stage 3 radio work changes the stack story.